



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	The company experienced a Denial of Service (DoS) attack in which a malicious actor flooded the internal network with ICMP packets through an unconfigured firewall . The excessive traffic disrupted network operations for approximately two hours , preventing employees from accessing internal network resources. The incident response team blocked incoming ICMP traffic, shut down non-critical services, and restored critical business services. The investigation confirmed that the attack exploited missing firewall configurations.
Identify	The organization identified that the primary vulnerability was an unconfigured firewall that allowed unrestricted ICMP traffic into the network. Risk assessments and firewall configuration audits should be conducted regularly to identify misconfigurations. Critical assets such as internal network services, firewall configurations, servers, and network infrastructure should be inventoried and protected. Periodic security assessments and vulnerability scans should also be implemented.
Protect	To reduce future risk, the company implemented a firewall rule to rate-limit incoming ICMP packets, enabled source IP verification to detect spoofed IP addresses, and deployed an Intrusion Detection/Prevention System (IDS/IPS) to filter suspicious ICMP traffic. Additional protections

	include maintaining secure firewall configurations, regularly updating security devices, applying the principle of least privilege, training employees on cybersecurity awareness, and documenting security policies.
Detect	The company deployed network monitoring software to continuously monitor network traffic and detect abnormal ICMP traffic patterns. IDS/IPS systems now generate alerts whenever suspicious traffic exceeds normal thresholds. Security logs should be monitored continuously using centralized monitoring tools (such as a SIEM) to improve early detection of denial-of-service attempts and other network anomalies.
Respond	During the attack, the incident response team immediately blocked incoming ICMP packets , isolated the affected network traffic, temporarily shut down non-critical services, and restored critical network operations. After containing the incident, the cybersecurity team investigated the root cause, confirmed the firewall misconfiguration, documented the incident, implemented new firewall rules, and updated response procedures to improve future incident handling.
Recover	Critical network services were restored after the malicious traffic was blocked. Firewall configurations were corrected, IDS/IPS and monitoring tools were deployed, and new security controls were implemented to prevent similar attacks. The organization should continue monitoring the network, verify that all services operate normally, perform post-incident reviews, update incident response documentation, and conduct regular testing to validate the effectiveness of the new security measures.

Reflections/Notes:

This incident demonstrates that even a single firewall misconfiguration can allow a successful DoS attack. Applying the **NIST Cybersecurity Framework (Identify, Protect, Detect, Respond, Recover)** provides a structured approach for improving organizational cybersecurity. Continuous monitoring, regular security audits, properly configured firewalls, IDS/IPS deployment, employee awareness, and periodic penetration testing are essential to reducing future cybersecurity risks.